

# EV Battery Analysis Report

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## Executive Summary

| Metric              | Value            |
|---------------------|------------------|
| Latest SoC          | 85.46%           |
| Latest SoH          | 95%              |
| Latest Voltage      | 386.09 V         |
| Maximum Temperature | 38.62 °C         |
| Battery Health      | Excellent        |
| Charging Duration   | 2.5 Hours        |
| Overall Status      | Normal Operation |

## Dataset Information

Number of Records: 8640

Start Time: 2026-05-15 00:00:00

End Time: 2026-05-15 23:59:50

## Battery Health Summary

Latest SoC: 85.46%

Minimum SoC: 80.09%

Maximum SoC: 85.49%

Latest SoH: 95%

Battery Health Status: Excellent

## Charging Analysis

Total Charging Duration: 2.5 hours

Charging Activity Status: Moderate charging activity

State of Charge Trend

State of Charge Over Time

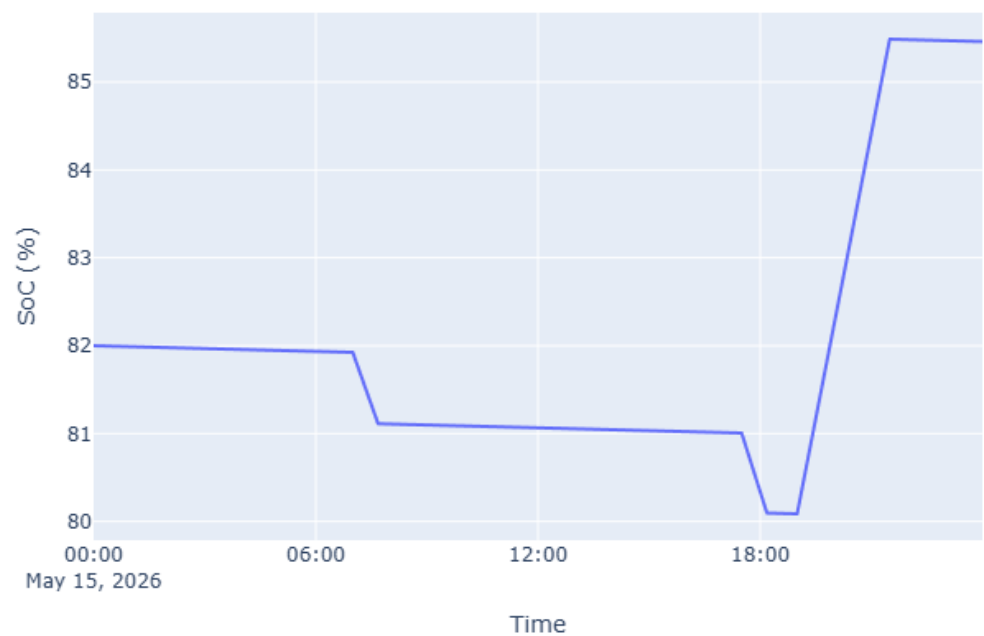


Figure 1: State of Charge variation over time.

## Voltage Analysis

Battery Voltage Over Time

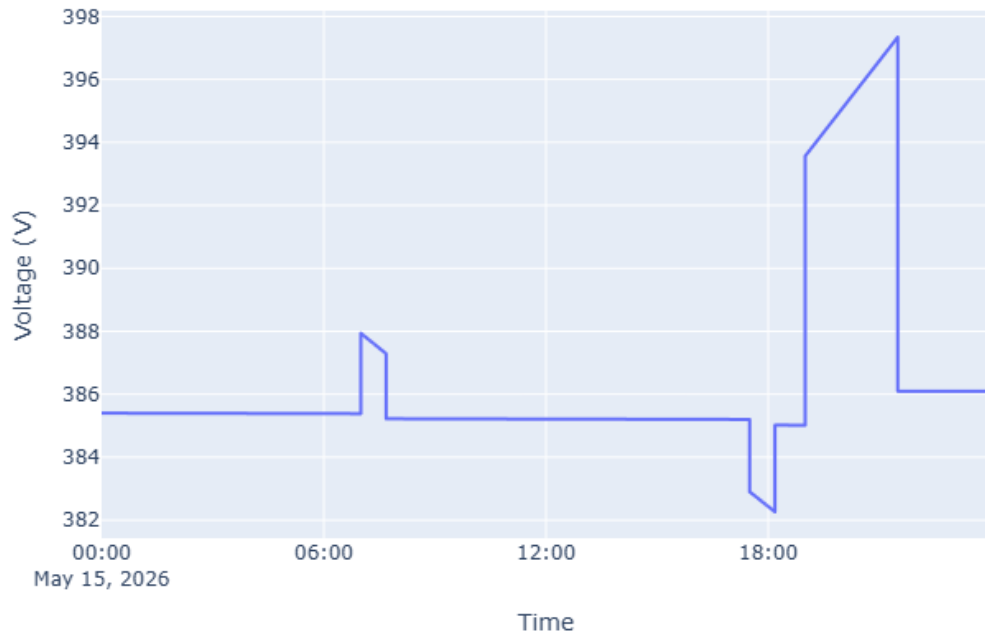


Figure 2: Battery pack voltage variation over time.

Voltage ranged from 382.27 V to 397.34 V.

## Current Analysis

Battery Current Over Time

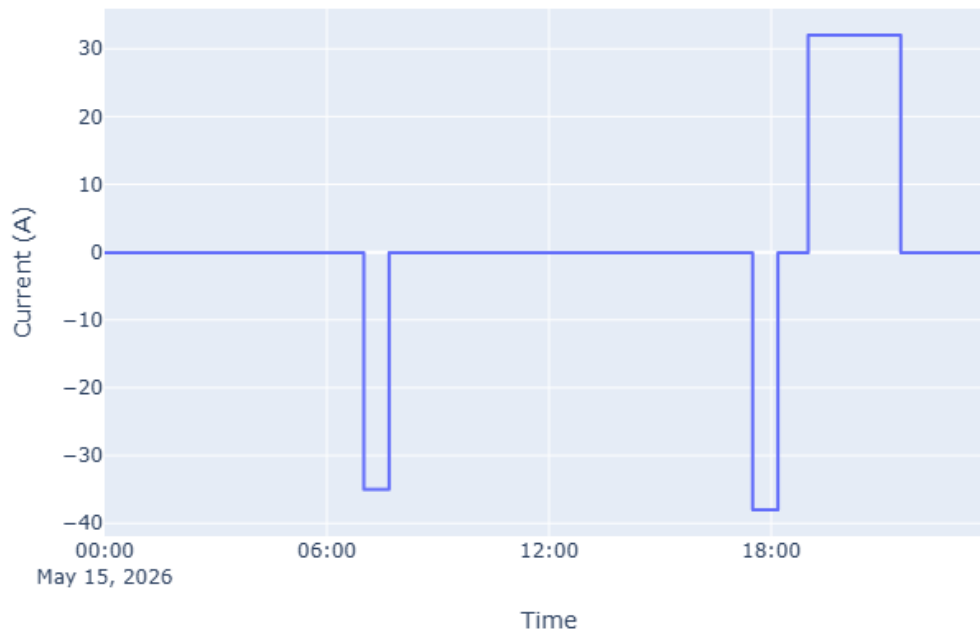


Figure 3: Battery pack current variation over time.

Maximum discharge current: 32.0 A

Maximum charging current: -38.0 A

## Temperature Analysis

Battery Temperature Over Time

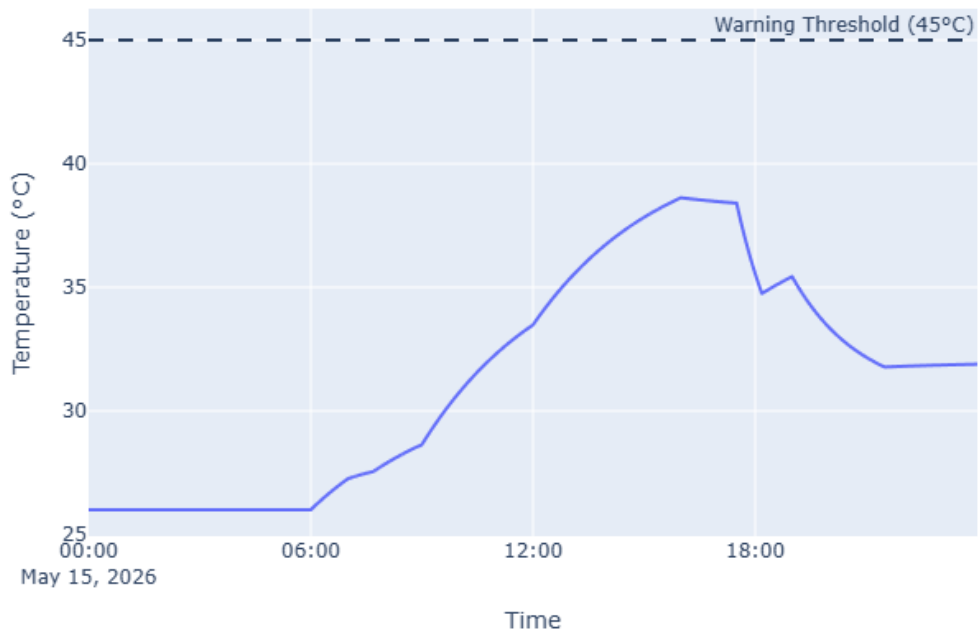


Figure 4: Battery pack temperature variation over time.

Maximum battery temperature recorded was 38.62 °C.

Temperature Status: Normal

### Alerts and Warnings

No critical alerts detected.

### Conclusion

The battery operated with a maximum temperature of 38.62°C, maintained a State of Health of 95%, and showed moderate charging activity. Overall battery condition was assessed as Excellent.